

Total No. of Questions : 8]

[Total No. of Printed Pages : 1

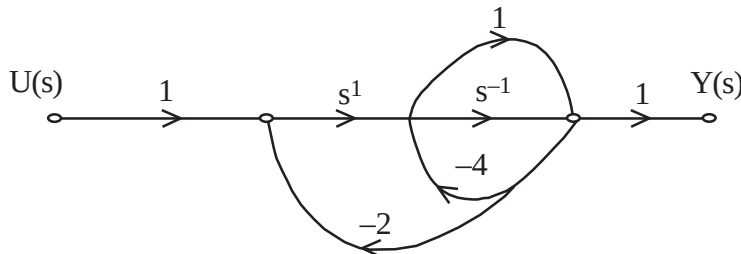
Roll No

EX-602 (GS)
B.E. VI Semester Examination, June 2020
Grading System (GS)
Control Systems
Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
 ii) All questions carry equal marks.

1. Draw the time response of a first order control system due to unit step, unit ramp and unit impulse function.
2. The signal flow graph for a system is given below. Find the transfer function $\frac{Y(s)}{U(s)}$



3. Explain phase lag and lead network used in compensation network.
4. What do you understand by modelling of electrical systems? Explain force voltage analogy with electrical circuit.
5. The open loop transfer function of a unity feedback control system is given by

$$G(s) = \frac{K}{(s+2)(s+4)(s^2+6s+2s)}$$

Discuss the stability of closed loop system as a function of K. Determine the values of K which will cause sustained oscillations in the closed loop system. What are the corresponding Oscillations of frequencies.

6. Explain the concept of Relative Stability and Absolute Stability.

OR

Write a short note on the advantages of Bode plot.

7. Characteristic equation of a system is given as:

$$S^6 + 2S^5 + 8S^4 + 12S^3 + 20S^2 + 16S + 16 = 0$$

Find the roots to ascertain if it is marginally stable or unstable.

- 8 Write a short note on any two of the following:

a) Standard Test Signals b) Poles and Zeros c) Power amplifier
